

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Introduction:**

The purpose of this project is to design and implement a data integration and aggregation process for customer and order data stored in CSV files. The goal is to load the data into Delta tables, perform data cleansing, create summary tables, and implement Slowly Changing Dimension (SCD) type 2 logic to track changes in customer data.

- Business Requirements:**

1. **Data Ingestion**

- Read source CSV data from a specified volume and load it into Delta tables for customer and order data.
- The file paths for customer and order data are
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`, respectively.

2. **Data Schema**

- The schema for the customer table is: `CustId`, `Name`, `EmailId`, `Region`.
- The schema for the order table is: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`.

3. **Data Cleansing**

- Remove "Null" or null records from both customer and order tables.
- Remove duplicate records from both customer and order tables.

4. **Order Summary Table Creation**

- Create an "ordersummary" table if it does not exist in the `gen\_ai\_poc\_databrickscoe` catalog and `sdlc\_wizard` schema.
- The schema for the "ordersummary" table is: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.

5. **Data Integration and SCD Type 2 Implementation**

- Join customer and order data using the `CustId` field.
- Load the joined data into an SCD type 2 table (`ordersummary`) under the `gen\_ai\_poc\_databrickscoe` catalog, `sdlc\_wizard` schema.
- Implement logic to update the SCD type 2 table whenever there is a change in the customer table.
- Old records should be made inactive, and new records should be made active.
- Update `StartDate` and `EndDate` accordingly.

6. ****Customer Aggregate Spend Table Creation****

- Create a "customeraggregatespend" table with columns "Name", "TotalAmount", and "Date" in the `gen\_ai\_poc\_databrickscoe` catalog and `sdlc\_wizard` schema if it does not exist.

7. ****Data Aggregation****

- Aggregate the "TotalAmount" column from the "ordersummary" table and group by "Name" and "Date" columns.
- Load the aggregated data into the "customeraggregatespend" table.

• **Functional Requirements:****

1. The system should be able to read CSV data from a specified volume.
2. The system should be able to load data into Delta tables.
3. The system should perform data cleansing (remove null/null records and duplicates).
4. The system should create the "ordersummary" table if it does not exist.
5. The system should implement SCD type 2 logic for the "ordersummary" table.
6. The system should create the "customeraggregatespend" table if it does not exist.
7. The system should aggregate data from the "ordersummary" table and load it into the "customeraggregatespend" table.

• **Non-Functional Requirements:****

1. The system should ensure data consistency and integrity.
2. The system should be able to handle large volumes of data.
3. The system should be scalable and flexible to accommodate changing business needs.

• **Assumptions and Dependencies:****

1. The input CSV files are in the correct format and are located at the specified paths.
2. The Delta tables are created with the correct schema.
3. The SCD type 2 logic is implemented correctly to track changes in customer data.

• **Acceptance Criteria:****

1. The system successfully ingests data from CSV files into Delta tables.
2. The system performs data cleansing and removes null/null records and duplicates.
3. The "ordersummary" table is created with the correct schema.

4. The SCD type 2 logic is implemented correctly for the "ordersummary" table.
5. The "customeraggregatespend" table is created with the correct schema.
6. The system aggregates data correctly from the "ordersummary" table and loads it into the "customeraggregatespend" table.

By following this BRD, the development team should be able to design and implement a data integration and aggregation process that meets the business requirements and is scalable, flexible, and maintainable.